



POLITECNICO
MILANO 1863

**MULTIMODAL INTERACTIVE
TECHNOLOGIES AND
APPLICATIONS**

Main Tasks for Heuristic Evaluation

Botanical Garden

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This document defines the three main interaction tasks used for heuristic evaluation of the Botanical Garden prototype. The tasks focus on how the system combines visual context, plant selection, speech or typed input, LLM support, local fallback logic, and adaptive information panels.

1 Task 1 — Visual Detection and Multimodal Disambiguation

Goal. The user wants to identify a plant using an ambiguous deictic question such as “*what is this*” or “*what plant am I looking at?*”. The task evaluates whether the system can detect that the reference is unclear and guide the user to the intended plant.

Modalities. *Input:* current camera view, speech-to-text, typed text, and click/touch selection on the A/B/C overlay. *Output:* candidate highlights in the 3D scene, labelled overlay options, information panel, and TTS feedback.

Flow.

1. The user asks an ambiguous identification question, for example “*what is this*” or “*what plant am I looking at?*”.
2. If no plant is already selected, the frontend checks the plants in the current visual field. When multiple visible plants are detected, the system enters a referential ambiguity flow. If visibility detection cannot determine a single clear target and the current scene contains multiple candidate plants, the scene candidates are used as a fallback ambiguity set.
3. The system generates A/B/C labels for the candidate plants, highlights them in the 3D scene, and displays a clarification overlay. A backend LLM call may be used to produce a natural clarification prompt; if this fails, a local prompt is used.
4. The user resolves the ambiguity by clicking one candidate, typing a letter, speaking a letter, or saying the plant name.
5. If the answer cannot be matched, the system asks the user to choose again. After repeated failures, it guides the user toward directly clicking the intended target.
6. Once a candidate is resolved, the selected plant is highlighted and its information card is shown.

Success criteria. Ambiguity is triggered when the user uses an unclear reference and more than one plant is visible or otherwise plausible in the current scene; every candidate is clearly labelled; clicking, typing, and voice-based letter selection work; failure cases provide a retry path and a direct-click fallback; no plant-specific information card appears before the target is resolved.

2 Task 2 — Plant Selection and In-depth Dialogue

Goal. The user selects one plant and asks targeted follow-up questions about its properties, such as basic botanical information or medicinal value. The task evaluates whether the system keeps the selected plant in context and presents the right type of information panel.

Modalities. *Input:* plant selection by click/raycast, speech or typed natural-language query, and fallback buttons for supported topics. *Output:* selected-plant highlight, botanical or medicinal information panel, fallback panel for unsupported interests, and TTS narration.

Flow.

1. The user selects a plant directly, or resolves Task 1 ambiguity so that one plant becomes the current selection.
2. The user asks a property question, for example “*what is its medicinal value?*”, “*tell me about this plant*”, or “*what is it used for?*”.
3. The frontend parses the intent and topic of interest. For medicinal value, medical, or herbal-use questions, the system opens the dedicated medicinal information panel.
4. For supported botanical queries, the system displays the general plant information card. Information generation first attempts to use the backend LLM endpoint; if generation fails or is disabled, the UI still falls back to structured local plant data.
5. If the user asks for an unsupported interest, such as a topic not available in the current prototype, the system shows a fallback interaction that guides the user toward supported topics such as botanical features or medicinal value.
6. Subsequent follow-up turns remain attached to the selected plant until the user selects another plant or closes the plant panel.

Success criteria. The selected plant remains visually highlighted; medicinal questions open the medicinal panel; general botanical questions open the standard information card; unsupported interests do not dead-end but offer supported alternatives; local fallback prevents the core interaction from breaking when LLM generation is unavailable.

3 Task 3 — Contextual Plant Comparison

Goal. The user compares the currently selected plant with another plant mentioned by name or inferred from visit history. For example:

`Is this more drought tolerant than the ficus?`

Flow.

1. The user has a current plant selected and asks a comparison question, such as “*Is this more drought tolerant than the ficus?*” or “*compare it with the previous plant*”.
2. The frontend parses the comparison intent and resolves the second plant. It first checks explicit plant names and aliases, then uses the user’s visited plant history for references such as “*the previous one*”. If the target was not visited but exists in the plant database, it can still be used as a global database fallback.
3. The comparison view displays both plant descriptions and key attributes, including drought tolerance, height, lifespan, and medicinal value where available.
4. The frontend sends both resolved plant records to the backend `/api/compare` endpoint when LLM comparison speech is enabled. The backend is responsible for generating a natural-language comparison answer, while the frontend remains responsible for rendering the structured side-by-side result.
5. If no valid comparison target can be resolved, the system explains the problem and asks the user to select another plant or use a known plant name.

Success criteria. The system resolves comparison targets from explicit names, aliases, recent history, or the global plant database; the comparison panel shows both plants in a structured layout; drought tolerance is compared clearly in the current prototype; missing targets produce a useful error instead of a silent failure; the spoken summary and visual comparison remain consistent.